

CENTRALIZED SMART ASSET MANAGEMENT FOR SRI LANKAN RAILWAYS (TRACK MIND)

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Acknowledgement

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Abstract

The Sri Lanka Railways Department presently maintains a huge stock of rolling stock spares, and maintenance tools, infrastructure assets in a largely paper-based ledger system. This disorganized strategy has caused huge operational inefficiencies such as asset pilferages, slow repair of the assets because of lost parts, and the inability to see the real-time flow in the regional divisions. TrackMind is a proposed project that suggests a complete digitalization of the railway inventory ecosystem with the help of a web-based Asset Management System.

The essence of TrackMind is to computerize the inventory cycle, including procurement, and disposal, and to introduce QR code integration, which allows to identify the asset in a matter of seconds due to the error-free process. The system architecture has been tailored to the complex hierarchy of the organization structure of the railways, and deals with inventory in many Divisions, Stations and Warehouses (including Workshops) in a centralized database. This will allow a single physical location to get a single available position of a Storekeeper whereas the Administrators have an overall picture of stock around the country.

One of the most significant innovations of this project is the combination of a privacy-oriented AI Assistant based on Retrieval-Augmented Generation (RAG). In contrast to typical keyword search engines, this module uses a self-written Context Loader to match the structured database entries with a Local Embedding Model so that non-technical employees can query stock levels with natural language (e.g., "Where can I find the brake pads to Class M4 engines?). Using MERN stack (MongoDB, Express, React, Node.js) as the backbone of the application and local vector embeddings to provide intelligence functionality, TrackMind provides scalable, cost-effective, and secure solution that can be entirely adapted to meet the particular modernization requirements of Sri Lanka Railways.

Keywords: *Asset & Inventory Management, Retrieval-Augmented Generation (RAG), MERN Stack, QR Code Tracking, Local Embeddings, Sri Lanka Railways.*

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List of Abbreviations

API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DB	Database
ERD	Entity-Relationship Diagram
GDPR	General Data Protection Regulation
GPS	Global Positioning System
HTML	HyperText Markup Language
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
ICT	Information and Communication Technology
IoT	Internet of Things
JS	JavaScript
JSON	JavaScript Object Notation
JWT	JSON Web Token
LLM	Large Language Model
MERN	MongoDB, Express.js, React.js, Node.js (Stack)
ML	Machine Learning
NLP	Natural Language Processing
PDF	Portable Document Format
QR	Quick Response (Code)
RAG	Retrieval-Augmented Generation
RBAC	Role-Based Access Control
RFID	Radio Frequency Identification
SLR	Sri Lanka Railways
SPA	Single Page Application
SQL	Structured Query Language
TLS	Transport Layer Security
UI	User Interface
UX	User Experience

1. Introduction

1.1 Background

The world is changing towards becoming digital in its railway sector due to the development of technologies in the areas of artificial intelligence, Internet of Things (IoT), cloud analytics, and machine learning. It is estimated that smart railways market will expand USD 36.5 billion in 2025 and USD 54.3 billion in 2030 as a result of significant investments in modernization in Asia, Europe, and North America. Current railway networks around the world use AI-driven predictive maintenance, virtual assets under observation through digital twins, real-time tracking of the condition with a network of IoT sensors, and complex analytics platforms to streamline the processes.

As an example, European high-speed rail networks and major Asian railway companies use a digital twin with AI analytics to track the state of equipment, forecast that it needs maintenance before it goes wrong, and find out the wear patterns of rail infrastructure in its initial stages. Research has shown that AI-based predictive maintenance can decrease equipment downtime by about 20 percent of what would otherwise be the case with reactive maintenance practices, and has more than 90 percent accuracy in detecting defects. Moreover, international rail operators, including (Netherlands), Indian Railways and Chinese railways applied successfully to RFID (Radio Frequency Identification) technology to gain automated rolling stock tracking, and allow real-time viewing of the asset position, state, and repair condition on the scattered railway networks.

In the context of Sri Lanka, the Department of Railways has been partially digitized with the use of e-ticket systems (including the online booking platform called Pravesha.lk and the mobile QR-code ticketing) in order to cover passenger-facing services and minimize the loss of revenues in terms of ticketless transport. Also, the continual improvements of telecommunications networks being financed by the Asian Development Bank are an indication that the government is determined to upgrade the railway infrastructure. Nevertheless, these online projects have not been replicated to the important backend role of asset and inventory management. The internal asset tracking, spares and equipment maintenance records are still predominantly manual and are maintained in disaggregated paper registers, spreadsheets and isolated department databases in the various workshops, stations and depots. This is a systemic failure, even though there are digital solutions to this scenario, proven to be effective in other

large railway networks, which underscores the major limitation on the effectiveness of operations and asset responsibility in Sri Lanka Railways.

1.2 Problem Statement

Sri Lanka Railways (SLR) currently manages its extensive network of nearly 400 stations and 12 depots without a centralized, integrated system for its physical assets. Instead, critical data regarding rolling stock and inventory is fragmented across manual paper registers and isolated Excel spreadsheets, creating information silos that prevent real-time visibility into asset availability. This lack of coordination has forced the department into an inefficient reactive maintenance model, directly contributing to acute operational failures. Consequently, despite a USD 90 million investment in new rolling stock, 331 locomotives and 131 power sets remain idle due to maintenance delays and spare parts unavailability, causing over 10,000 service cancellations in recent years and a 72% rate of late arrivals in 2022.

Beyond operational delays, this disjointed framework has resulted in significant financial losses and a breakdown in governance. The inability to verify inventory usage has led to documented losses of Rs. 25.3 million due to misuse and improper asset disposal, alongside the discovery of 69 abandoned wagons and 132 unoccupied rail quarters lacking basic ownership records. Furthermore, this manual ecosystem has paralyzed internal accountability; audit committees failed to meet for five consecutive years (2019–2024), leaving 27 major queries unanswered. These conditions directly undermine the government's mandate for digital transparency as outlined in the Public Assets Management Bill (2022) and the 2026 National Digitalization Strategy.

1.3 Motivation & Significance of Project

TrackMind helps to directly satisfy a vital and unsatisfied requirement in the SLR by implementing international best practices to the area that was prioritized poorly before. As the department has performed exceptionally in digitalizing passenger-facing service like online ticketing, the internal control of tangible resources and spares has been predigital. This disjuncture has not arisen because of the unavailability of technology but has been a result of historically neglecting the operations of the back office. TrackMind is a strategic intervention that can address this particular inefficiency to bring about modernization of the activities that form the fabric of railway operations.

The need to come up with this system is motivated by the following four factors

1. **Sealing a Verified Operational Loophole:** According to past audit reports conducted by the National Audit Office, inefficiencies have been found in the use of assets and there was no centralized supervision at SLR. These are not some hypothetical issues but rationalized operational failures resulting in financial loss. It is the motivation of TrackMind to overcome these particular, documented shortcomings with an integrated digital solution.
2. **Conformity with National Policy:** The project will be in line with the 2026 Sri Lankan Digitalization Strategy and the particular requirements of the Public Assets Management Bill. TrackMind helps the national goal of developing data-driven governance structures of public property by developing a centralized digital registry of railway assets.
3. **Lessening of Quantifiable Operational Costs:** There are real economic gains that come as a result of using AI in tracking the assets. The case studies carried out all over the world show that predictive maintenance and real time inventory tracking can cut down up to 20 percent of equipment downtime. TrackMind seeks to implement such efficiencies at SLR, reducing the unnecessary procurement process, avoiding the theft of assets, and expediting the execution of audits.
4. **Democratizing access to data through AI:** It is a fundamental idea to enable non-technical workers in the field to access complex data. The system enables storekeepers and mechanics to make decisions based on data without training them to be sophisticated at IT by using Retrieval-Augmented Generation (RAG) to make queries in natural language (e.g., Which spare parts are below minimum stock?).

1.4 Aims and Objectives

1.4.1 Aims

The main objective of the project is to design and develop TrackMind, which is a centralized Asset and Inventory Management System based on the Sri Lanka Railways. The system will seek to substitute manual ledger processes with a secure digital ecosystem that involves the use of the QR code technology to track processes in real-time and the Artificial Intelligence (RAG)

to support the natural language data search to improve transparency and accountability in the operations.

1.4.2 Objectives

- To integrate the scattered inventory data sources through building a single MongoDB database architecture. This will be based on the fact that the system manages to bring together data of at least 3 different administrative levels (Division, Station, Warehouse) into one single dashboard that will be considered the Truth.
- To speed up the process of identifying assets with the addition of a mobile-responsive QR Code scanning module. The measure of success will be the decrease of the average speed of the manual search to less than 5 seconds per scan in order to identify assets.
- To democratize data access through a Retrieval-Augmented Generation (RAG) engine, which is a natural language processing engine. This will be quantified by how fast the system will respond to inventory queries (e.g., "Stock levels for M4 Engine") without requiring any db query input by some user.
- To impose high accountability through incorporation of Role-Based Access Control (RBAC). This goal is met as a system will create a 100 percent traceable digital audit trail of all stock transactions (Issue/Receive) so that no movement of assets will be unrecorded.
- To support real time analytics dashboard with a view of making data-driven decisions. This will be calculated by the system's ability to automatically detect and notify the management on the assets that are idle beyond 30 days.

2. Literature Review

Asset and inventory management is a very important application in the sphere of railway administration, which guarantees safety and efficiency. This data is a discussion of the current solutions that are being used in the industry and the world at large in order to determine the gaps that the proposed TrackMind system is seeking to fill. The review paper has a background analysis of the current practices in Sri Lanka Railways (SLR) and a critical analysis of five key systems starting with traditional manual systems up to the state-of-the-art enterprise solutions.

2.1 Critical evaluation of 5 existing solutions

Five different systems, including the current manual system adopted by SLR, generic spreadsheet software, the existing digital ticketing platform (Pravesha.lk) adopted by SLR, global Enterprise resource Planning (ERP) solutions (SAP), and logistics platforms based on IoT were chosen to analyze the positioning of TrackMind.

2.1.1 Manual Paper-Based Recording (Current SLR Standard)

This is the legacy system currently in operation across most SLR stations and workshops. It relies on physical logbooks, paper registers, and filing cabinets to record asset details, maintenance history, and inventory movements. [1]

System Components: Hardcopy ledgers, paper and manual filing storage.

Advantages:

- **Zero Implementation Cost:** There will be no money invested in digital infrastructure or hardware.
- **Institutional Knowledge:** Years of institutional experience imply that there is no urgent need to train the personnel.

Disadvantages:

- **Information Latency:** Retrieval of information is slow and manual. It may require days to know the history of the service of a particular component and this does not allow quick decision taking.

- **Absence of Analytics:** The paper is not dynamic, thus making it impossible to analyze any trends or do predictive stock planning (reactive maintenance).
- **Vulnerability:** Records have the potential to be physically damaged, lost, and altered unlawfully without an electronic audit trail.

2.1.2 General Spreadsheet Software

Some of the administrative units at SLR have shifted to the use of spreadsheet software to computerize records. Although this is better than paper, it is a form of digitization but not digitalization. [2]

Advantages:

- **Accessibility:** Due to the low learning curve, the administrative staffs are able to carry out simple data sorting and filtering without any special training in technical skills.
- **Cost-Efficiency:** Uses the available office software licenses.

Disadvantages:

- **Data Fragmentation:** The problem with the lack of a central database is that there are broken down data silos in the data which causes conflicting records of inventory to exist in various stations.
- **Concurrency Problems:** Files are not normally multi-user so real time cooperation or multiple warehouses cannot be updated at the same time.
- **Security Risks:** Files also do not have a tough access control system that protects them against accidental deletion, duplication, or manipulation.

2.1.3 Pravesha.lk (SLR Smart Ticketing Platform)

The best existing digital initiative by SLR is "Pravesha.lk. Nevertheless, it has a domain specific architecture to passenger transit and not engineering. [3]

Advantages:

- Established Digital Adoption: Confirms that SLR workforce and infrastructure can accommodate a successful transmission to QR-code-based workflows.
- Revenue Integrity: Leakage of revenue was reduced successfully via real-time online monitoring. [4]

Disadvantages:

- Domain Restriction: The system is designed to support ephemeral transactions (tickets) only and makes no facilities to support the lifecycle of persistent assets (depreciation, maintenance records, re-stocking).
- Unmatched Logic: Passenger flow logic is completely dissimilar to supply chain warehousing logic, and the software cannot be used directly.

2.1.4 SAP + IBM Watson

SAP S/4HANA is a Tier-1 ERP used globally for finance, logistics, and asset management. SAP announced integration of IBM Watson AI to add natural language queries and predictive insights into SAP applications, including S/4HANA and related asset modules [5]

Advantages:

- Natural Language Interaction: Validates the TrackMind concept that non-technical users benefit most from conversational AI interfaces (e.g., asking questions in plain English).
- Holistic Integration: Seamlessly connects inventory data with procurement, finance, and HR modules.

Disadvantages:

- Prohibitive Cost: The licensing and implementation costs are exponentially higher than the budget for a focused internal department project.
- Operational Complexity: These generic enterprise systems require extensive customization and specialized training, often proving too complex for ground-level storekeepers. [6]

2.1.5 Roambee (IoT-Based Asset Tracking)

Roambee is a prominent industry solution for logistics, utilizing Internet of Things (IoT) sensors to provide real-time visibility of assets in transit. [7]

Advantages:

- Real-Time Telemetry: Excellent for tracking high-value moving assets (e.g., locomotives) across the rail network via GPS.
- Automation: Removing the need for manual scanning reduces human error to near zero.

Disadvantages:

- Hardware Dependency: Requires expensive active sensors for every tracked item. TrackMind's proposed passive QR code solution is significantly more scalable for high-volume, low-cost inventory (e.g., spare parts).
- Logistics Focus: Optimized for "Chain of Custody" in shipping rather than the complex "Warehousing and Maintenance" operations required by SLR workshops.

Table 1: Comparison of Existing Systems

System	Primary Domain	Centralized DB	Real-Time?	AI / NLP?	Tracking Method	Cost	User Skill
Manual / Paper	Record Keeping	No	No	No	Written Labels	Low	Low
Excel Sheets	Ad-hoc Data	No	No	No	Text ID	Low	Medium
Pravesha.lk	Passenger Tix	Yes	Yes	No	QR	Medium	Medium
SAP + Watson	Enterprise ERP	Yes	Yes	Yes	RFID / Barcode	V. High	High
Roambee	Logistics/ IoT	Yes	Yes	No	Active Sensors	High	High
TrackMind	Asset Management	Yes	Yes	Yes	QR Codes	Medium	Low

3. Project Plan and Initial Design

3.1 Components of the System

The TrackMind system architecture is composed of three integrated layers: a software stack for user interaction and logic, a database layer for persistent storage, and a hardware interface for physical asset tracking.

3.1.1 Software Component and Tasks

The system is developed as a web-based application utilizing the MERN Stack (MongoDB, Express.js, React.js, Node.js), ensuring cross-platform compatibility and scalability. The software components are divided into the Frontend Client, Backend API, and the AI Processing Module.

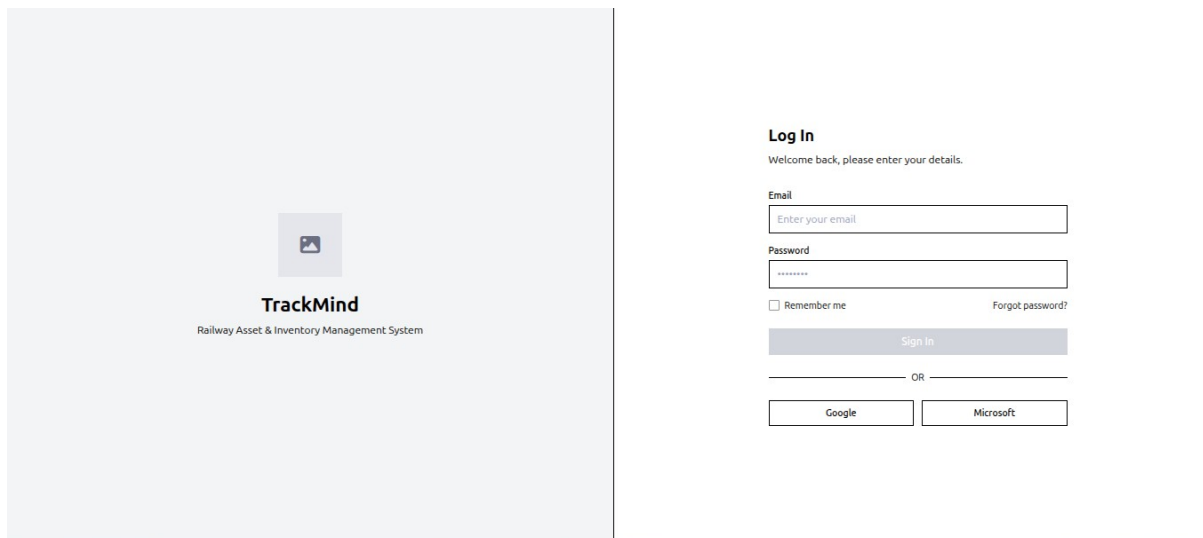
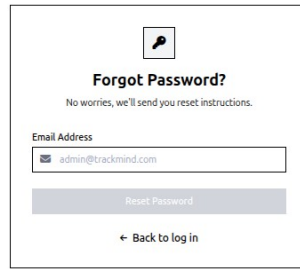


Figure 1: Login Screen Wireframe

The secure entry point to the TrackMind system. This interface enforces Role-Based Access Control (RBAC) by validating user credentials against the encrypted MongoDB database. Upon successful authentication, the system issues a JSON Web Token (JWT) to manage the user's session state securely.



Forgot Password?
No worries, we'll send you reset instructions.

Email Address

Reset Password

← Back to log in

Figure 2: Forgot Password Wireframe

This self service recovery module reduces the administration overhead as the user can securely reset their credentials. The workflow is also connected to an email service to send verification codes through which account access can be restored in a short time without security procedures being violated.



TrackMind
Dashboard Overview

Total Asset Value: **LKR 24,500,000** | Last Stock Alert: **24 Alerts** | Pending Requests: **12 Orders**

Inventory Management

AI Assistant
Powered by RAG Intelligence

Hello! I'm your AI assistant. I can help you with inventory queries, safety procedures, and asset information. How can I assist you today?

Ref: Safety_Manual.pdf

What is the reorder level for signal lamps?

According to the inventory guidelines, the reorder level for signal lamps (SKU: SIG-LAMP-001) is set at 50 units. When stock falls below this threshold, an automatic purchase request should be initiated.

Ref: Inventory_Guidelines.pdf
Ref: Signal_Equipment.pdf

Type your question...

AI responses are based on indexed documents

Figure 3: Dashboard AI chatbot Screen wireframe

The control center of the administration, which also shows the Key Performance Indicators (KPIs) in real time, including the total asset value and stock alerts. It has an interface with RAG-driven AI Assistant (bottom right) that enables non-technical personnel to require more

complicated inventory data in the form of natural language queries (e.g., Show me low stock in Galle).

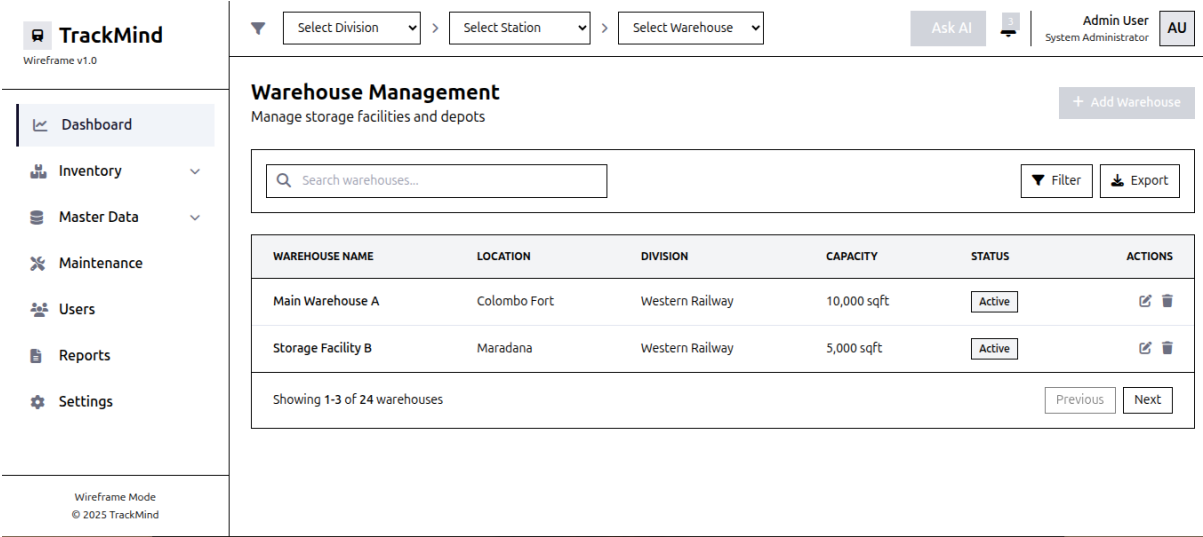


Figure 4: Warehouse Management Wireframe

This administrative interface enables configuration of physical storage location in the system. It allows connecting the warehouse entities and modifying them, associating them with particular railway stations to keep the appropriate Division → Station → Warehouse hierarchy.

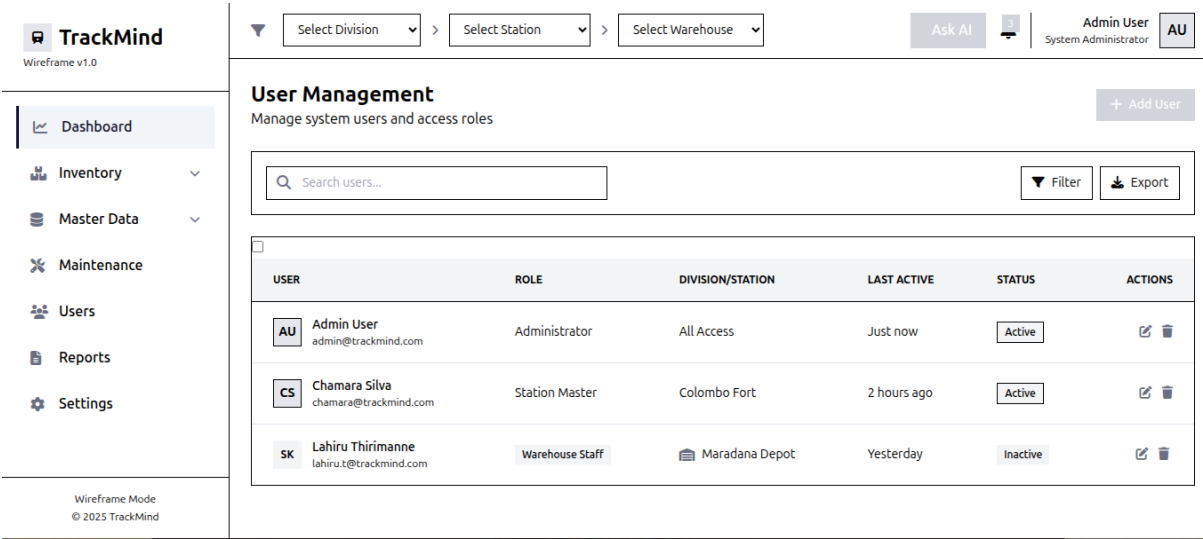


Figure 5: User Management Wireframe

Limited module where only the System Administrators use it to manage the workforce. It provides us with the opportunity to create new accounts of users, assign them with definite roles

(e.g., Storekeeper, Auditor, Manager), and make sure that sensitive functions are only shown to the authorized personnel.

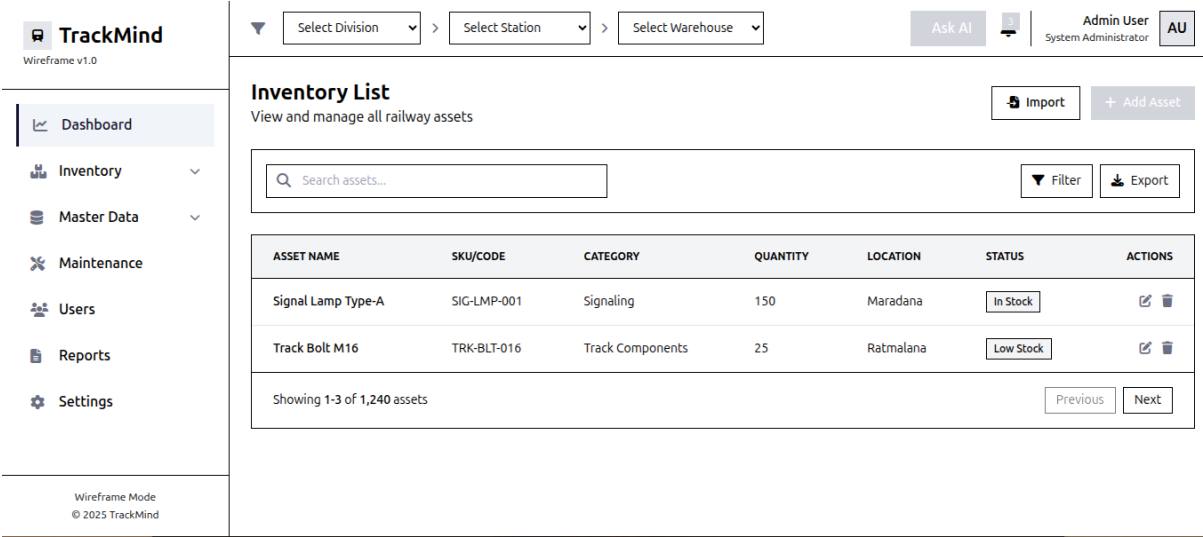


Figure 6: Inventory List Wireframe

The main operational perspective of Storekeepers and showing a table registry of all the rolling stock spares and tools. It has a high level of filtering, sorting, and coloured status display (e.g., Red: Low Stock). It is as well the catapult of QR code scanning activities in issuing or receiving things.

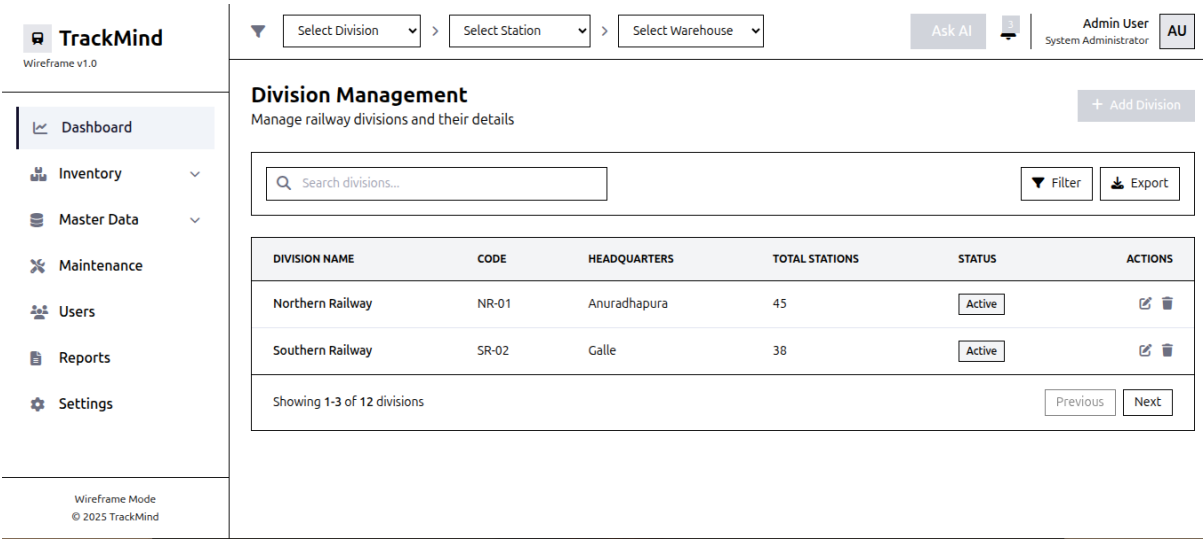


Figure 7: Division Management Wireframe

This is a screen that controls the top tier of the Sri Lanka Railways organizational structure. It enables administrators to subdivide stations into region Divisions (e.g., Western, Central, Northern), and the system provides an opportunity to create region-specific reports and to separate the inventory data in a logical way.

3.1.2 Database Component

To address the problem of "information silos" and support the hierarchical nature of Sri Lanka Railways, TrackMind utilizes a document-oriented NoSQL database.

- **Database Selection:** MongoDB was selected over traditional SQL databases due to its flexibility in handling semi-structured data (like varying asset specifications) and its high scalability for real-time inventory updates.
- **Database Schema Design:** The database structure follows the administrative hierarchy of the railway department. The schematic design is centered around the following key entities (as illustrated in the ERD in Appendix 5.3)
 - **Users:** Stores system access credentials. It includes fields for `user_id`, `password_hash` (encrypted), `role` (Admin, Storekeeper, Auditor), and `assigned_warehouse_id` to enforce data security.
 - **Warehouses:** Represents the physical storage locations. It contains attributes like `warehouse_id`, `location_name` (e.g., "Galle Workshop"), and is linked to parent Stations and Divisions.
 - **Products:** Acts as the master catalog. It defines static details such as `product_name` (e.g., "Brake Shoe Type-B"), `category`, `manufacturer` and `reorder_threshold`.
 - **Stock:** This is a dynamic entity that links a Product to a specific Warehouse. It tracks the `current_quantity_on_hand` and `last_updated` timestamp. This separation allows the same product to have different stock levels across different stations.
 - **Stock Transaction:** The most critical entity for governance. It serves as an immutable ledger recording every single inventory change. Attributes

include transaction_id, type (Issue/Receive/Adjust), user_id (who did it), timestamp, and related_job_card.

3.2 Proposed Methodology.

The TrackMind is developed in Agile Iterative Methodology, which offers continuously operating feedback and improvement. The system operation methodology is created in such a way that it removes the manual work by automating and using AI to aid in the process.

1. Operational Workflow

- Input: The Storekeeper logs on to the system when a new load of spare parts comes to the store. They no longer need to write in a ledger, they choose Get Goods and scan the barcode of the manufacturer or give the code of a new QR. Processing The React
- (Frontend) authenticates the input and transfers a secure API call to the Node/Express (Backend). The backend makes an update of the Stock collection in MongoDB and at the same time a Stock Transaction record.
- Output: the dashboard is updated immediately on the whole system. The alert of Low Stock is automatically activated when the quantity is less than the threshold reported on the master product record.

2. AI-Driven Inquiry (RAG Pipeline):

The system uses Retrieval-Augmented Generation (RAG) pipeline to address the problem of solving complex data retrieval. When the manager questions about enough signal cables to fix the Kandy line, the system does not necessarily go by the training information of the LLM.

- Step A: The system translates a query of the user in natural language to a database search vector.
- Step B: It retrieves the live stock information of the MongoDB database (Retrieval).
- Step C: It inputs this particular data as context to the LLM.
- Step D: This step uses the facts that have been retrieved to create a natural language response (Generation) using only the facts having been recalled to guarantee accuracy and avoid hallucinations.

3.3 Technologies Adapted

TrackMind system is developed on a strong and contemporary web architecture to provide scale and performance. The frontend interface is built with React.js that facilitates a quick and dynamic Single Page Application (SPA) platform that enables storekeepers to navigate the inventory without having to refresh the page. In the case of user interface design, Tailwind CSS is used both to maintain a consistent and contemporary look and to be able to develop a UI within a short period of time.

The backend API on the server side is implemented using Node.js and Express because these platforms have a non-blocking I/O model that effectively manages multiple asynchronous requests, and this is vital in a multi-user railway setting. MongoDB provides data persistence, which is a cloud-based NoSQL database that is designed specifically to store complex, hierarchical asset logs and unstructured maintenance data.

The system incorporates OpenAI/Gemini API to provide smart functions, and uses Large Language Models (LLM) as the driving force of the natural language Chatbot assistant. Security is the most important and JWT (JSON Web Tokens) is put in place to handle stateless user sessions and safeguard protected API routes. Lastly, all the life cycle development is handled through Git and GitHub, with strict version control and proper management of the source code.

3.4 Test and Deployment Plan

A stringent testing and deployment plan will be implemented to have a system that has sufficient reliability to perform critical railway operations.

Testing Strategy

1. **Unit Testing:** Each separate component will be tested individually (e.g., the function with the name Calculate Depreciation or the validation of the function Login) under the frameworks such as Jest to verify the logic is correct.
2. **Integration Testing:** The communication between the Backend API and the Database will be tested to make sure that stock updates are properly committed and that no errors are experienced generating transaction logs.

3. User Acceptance Testing (UAT): The prototype will be used by a trial group of real SLR employees (Storekeepers and Engineers) in a simulated setting. They will complete real life tasks (e.g., "Issue 50 brake pads) and make sure that the workflow corresponds to their physical actions.

Deployment Plan

1. Cloud Hosting: The pilot version will be hosted on a cloud service (e.g., AWS EC2 and vercel) to enable supervisors and examiners to access the pilot version remotely throughout the evaluation phase.
2. Maintenance and Monitoring: After the deployment, the application will record the system errors with the help of logging tools (such as Winston). The automatic backup of the database will occur after every 24 hours to avoid the loss of the data.

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5. Appendices

5.1 Timeline - Gantt chart

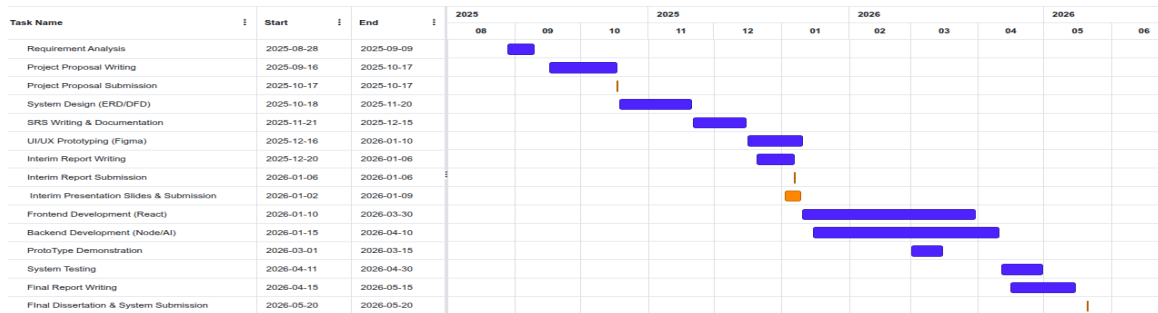


Figure 8: Gantt Chart

5.2 Data Flow Diagram

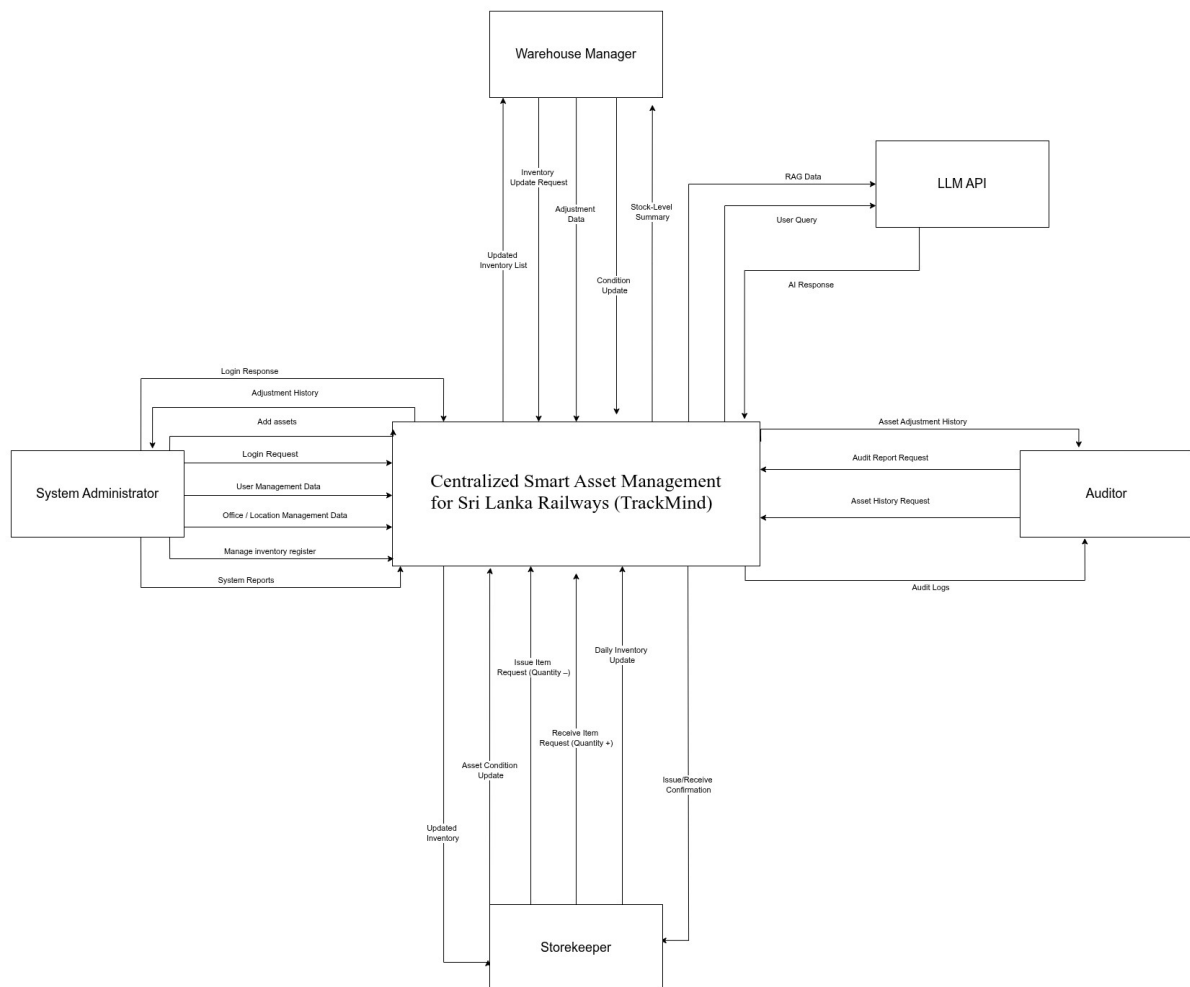


Figure 9: Level 0 Data Flow Diagram

The Level 0 Data Flow Diagram illustrates the high-level system boundaries and the information exchange between the TrackMind central process and its external environment. At the core of the system is the "Centralized Smart Asset Management" process, which acts as the primary integration hub for all inventory operations. This central process interacts with five distinct entities to maintain data integrity and facilitate operations. The System Administrator initiates the system's structural setup by providing "User Management Data" and "Office/Location Management Data," receiving "System Reports" in return to monitor overall health. Operational workflows are primarily driven by the Storekeeper, who sends "Issue" and "Receive" item requests to modify stock levels; in response, the system provides "Daily Inventory Updates" and confirmation messages to ensure physical stock aligns with digital records.

Parallel to daily operations, the Warehouse Manager and Auditor entities ensure oversight and compliance. The Warehouse Manager inputs "Adjustment Data" and "Condition Updates" to correct discrepancies, receiving "Stock-Level Summaries" and updated inventory lists to aid in procurement planning. The Auditor, designed as a read-only entity for security purposes, requests "Asset History" and "Audit Reports," to which the system responds with immutable "Audit Logs" that trace the lifecycle of railway assets. Finally, the system incorporates modern AI capabilities by interfacing with an external LLM API. The system packages user queries with relevant "RAG Data" (retrieved inventory context) and transmits them to the API, which returns an "AI Response" that is displayed to the user.

5.3 ER Diagram

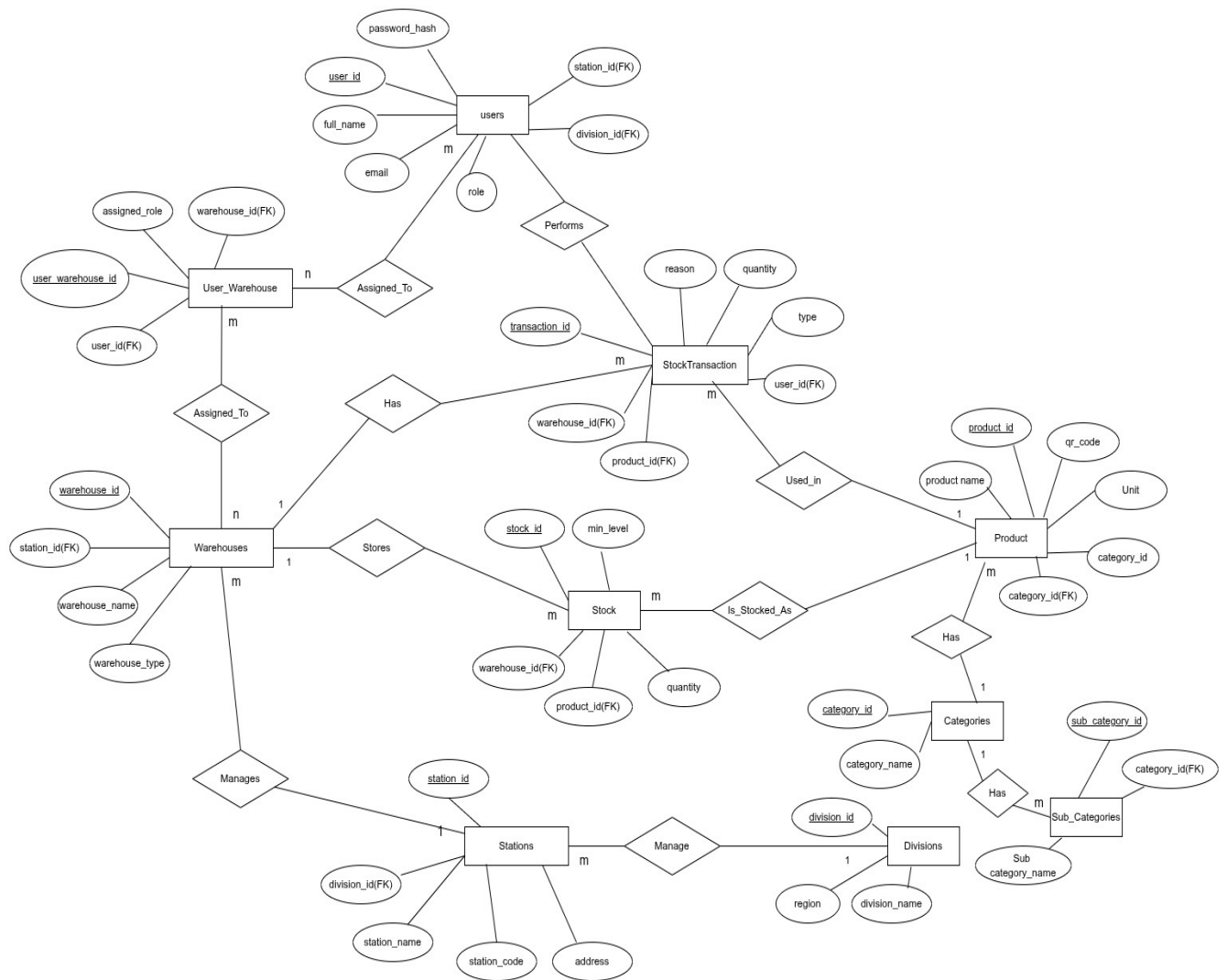


Figure 10: ER Diagram

The Entity-Relationship Diagram (ERD) depicts the normalized database schema designed to support the complex, hierarchical structure of SLR. The data model is built upon a strict location hierarchy where Divisions manage multiple Stations, and each Station manages specific Warehouses, allowing for scalable data aggregation at any administrative level. Security is enforced at the database level through the `User_Warehouse` intersection entity, which links users to specific warehouse locations; this ensures that staff can only modify stock within their assigned jurisdiction.

A critical architectural decision in the ERD is the separation of inventory data into Product and Stock entities. The Product entity stores static definitions such as the product name, unit type, and QR code, while the Stock entity tracks the dynamic quantity and minimum stock levels for that product at a specific warehouse. This allows the same item (e.g., a brake pad) to be tracked across hundreds of locations independently. Furthermore, to satisfy audit requirements, the schema includes a StockTransaction entity that records every inventory movement, linking the acting user, the specific stock item, and the transaction reason, thereby creating an immutable audit trail for the Auditor role.

5.4 SRS

The full Software Requirements Specification (SRS) document has been uploaded digitally to preserve the formatting of large architectural diagrams and detailed requirement tables.

[SRS Document Link](#)

5.5 Use case Diagram

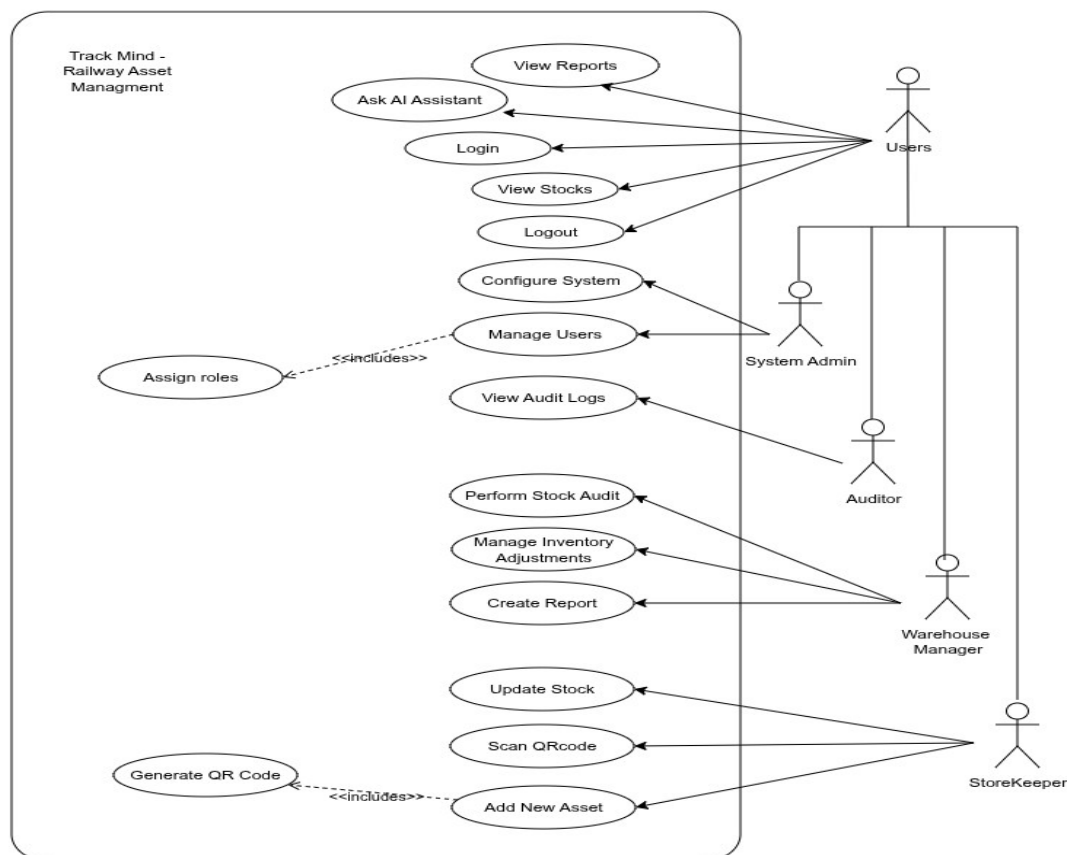


Figure 11: Use case Diagram

The Use Case Diagram defines the functional requirements of TrackMind and visually represents the Role-Based Access Control (RBAC) strategy employed by the system. The architecture is centered around a Generalization (Inheritance) relationship, where a base actor named "Users" represents the shared privileges of all railway staff. This design ensures that essential functions such as "Login," "Logout," "View Stocks," "View Reports," and "Ask AI Assistant" are universally accessible, promoting ease of use across the organization.

Specific operational roles are defined as specialized actors with distinct privileges. The System Admin holds exclusive rights to "Configure System" and "Manage Users," which includes the critical "Assign Roles" functionality to maintain system security. The StoreKeeper is focused on high-speed inventory tasks, utilizing "Scan QR Code" to identify assets and "Update Stock" to record daily transactions; notably, the "Add New Asset" use case automatically triggers "Generate QR Code," streamlining the registration process. Managerial oversight is handled by the Warehouse Manager, who can "Manage Inventory Adjustments" and "Create Reports," while the Auditor is restricted to the "View Audit Logs" and "Perform Stock Audit" use cases, ensuring a strict separation of duties for compliance.

Table 2: Use Case Summary Table

Use Case Name	Primary Actor(s)	Description
Login / Logout	<i>All Users</i>	Authenticates the user against the database and terminates the session securely to prevent unauthorized access.
Manage Users	System Admin	Allows the admin to create, update, or deactivate user accounts for new or departing railway staff.
Assign Roles	System Admin	<i>Included in Manage Users.</i> The specific action of granting permissions (e.g., assigning

		"Storekeeper" rights) to a user account.
Add New Asset	Storekeeper, Admin	The process of creating a new record for an item that has never been in the system before.
Generate QRCode	<i>System (Auto)</i>	<i>Included in Add New Asset.</i> The system automatically generates a unique QR code for every newly registered asset to facilitate future scanning.
Scan QR Code	Storekeeper	The act of using a QR code scanner or camera to instantly retrieve an asset's record or initiate a transaction.
Update Stock	Storekeeper	Modifying the quantity, status (e.g., "Good" to "Damaged"), or location of an asset in the database.
View Stocks	Storekeeper, Manager, Auditor	Allows users to browse the digital inventory, search for specific items, and view current stock levels across locations.
Create Report	Warehouse Manager	Generates a detailed, downloadable document (PDF/Excel) summarizing inventory status, movements, or valuation for a specific period.

View Reports	Manager, Auditor	Read-only access to per-generated dashboards and summaries for monitoring purposes.
Ask AI Assistant	Manager, Auditor, Admin	Enables the user to query the database using natural language (e.g., "Show me all critical low stock items in Galle") and receive an AI-generated answer based on real-time data.

5.6 Class Diagram

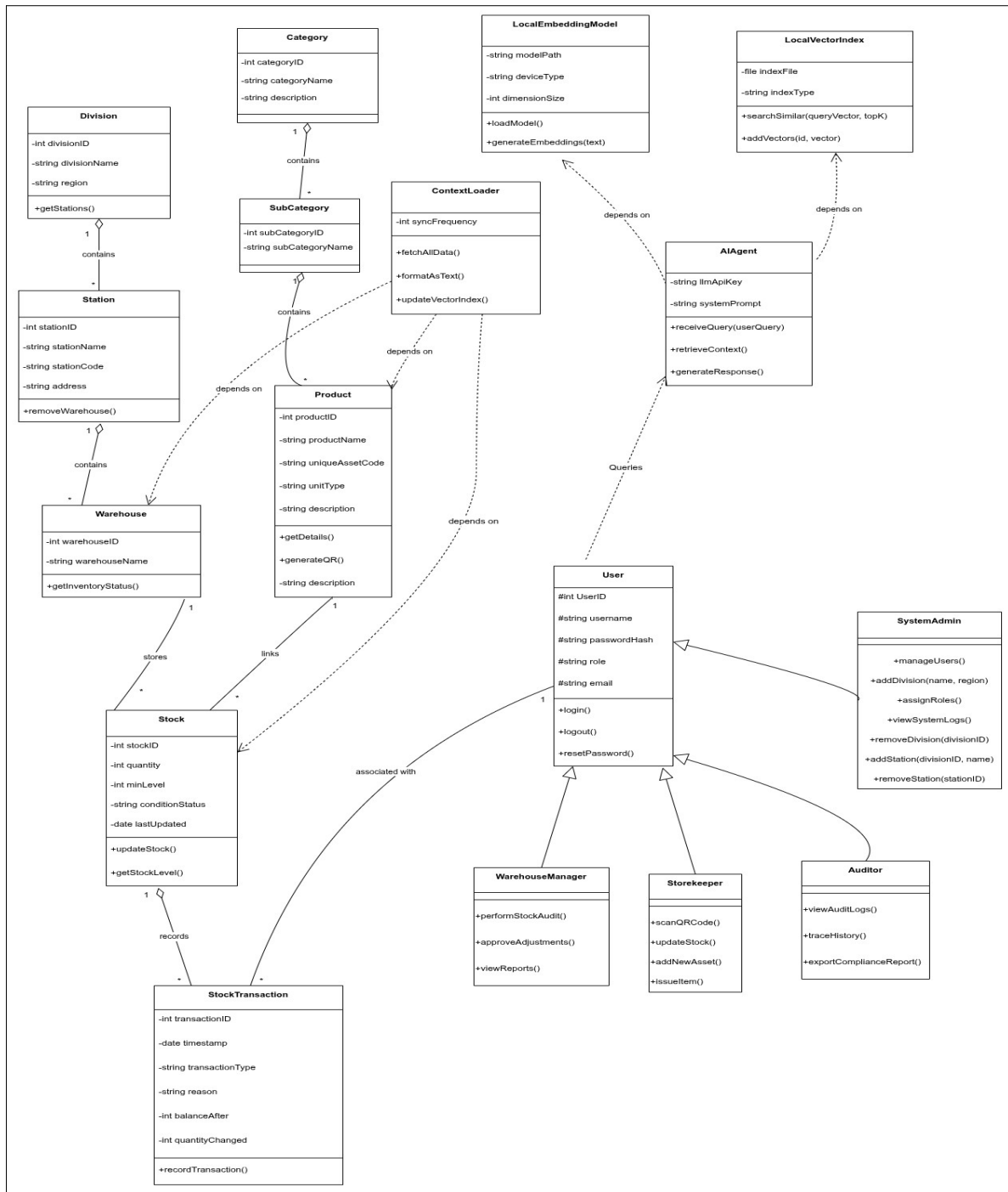


Figure 12: Class Diagram

The Class Diagram illustrates the Object-Oriented Design (OOD) of the TrackMind backend, highlighting the modular separation between core business logic and the AI subsystem. The system leverages polymorphism through a User parent class, which encapsulates secure session management methods like login() and logout(). This class is extended by specialized child classes SystemAdmin, WarehouseManager, Storekeeper, and Auditor—which implement role-

specific interfaces such as `scanQRCode()` or `viewAuditLogs()`, ensuring that code behavior aligns with the security policies defined in the Use Case model.

The diagram also details the architecture of the RAG engine. Unlike a standard CRUD application, TrackMind includes a dedicated `AI Agent` class that acts as the central controller for processing user queries. This agent depends on a `ContextLoader` to fetch real-time inventory data and a `LocalVectorIndex` to search for contextually similar information chunks. By separating the AI logic from the core `Product` and `Stock` classes, the system ensures that the intelligent assistant can be updated or modified without disrupting the primary inventory management functions.